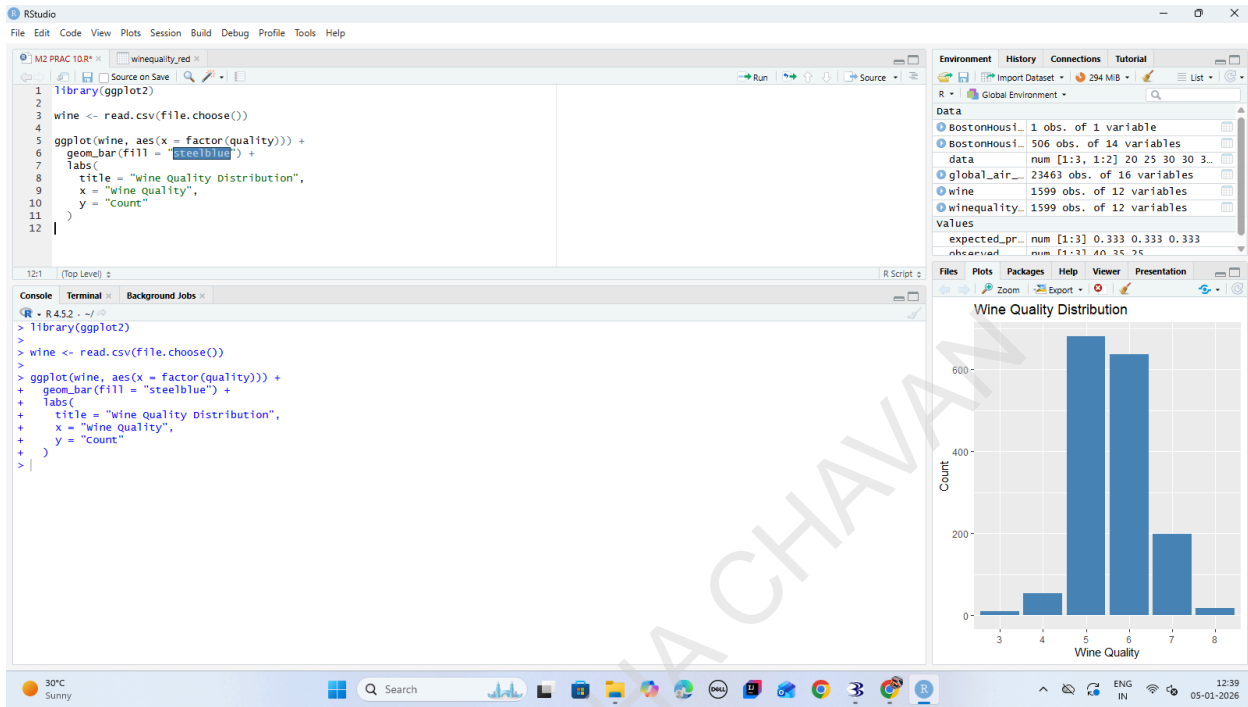


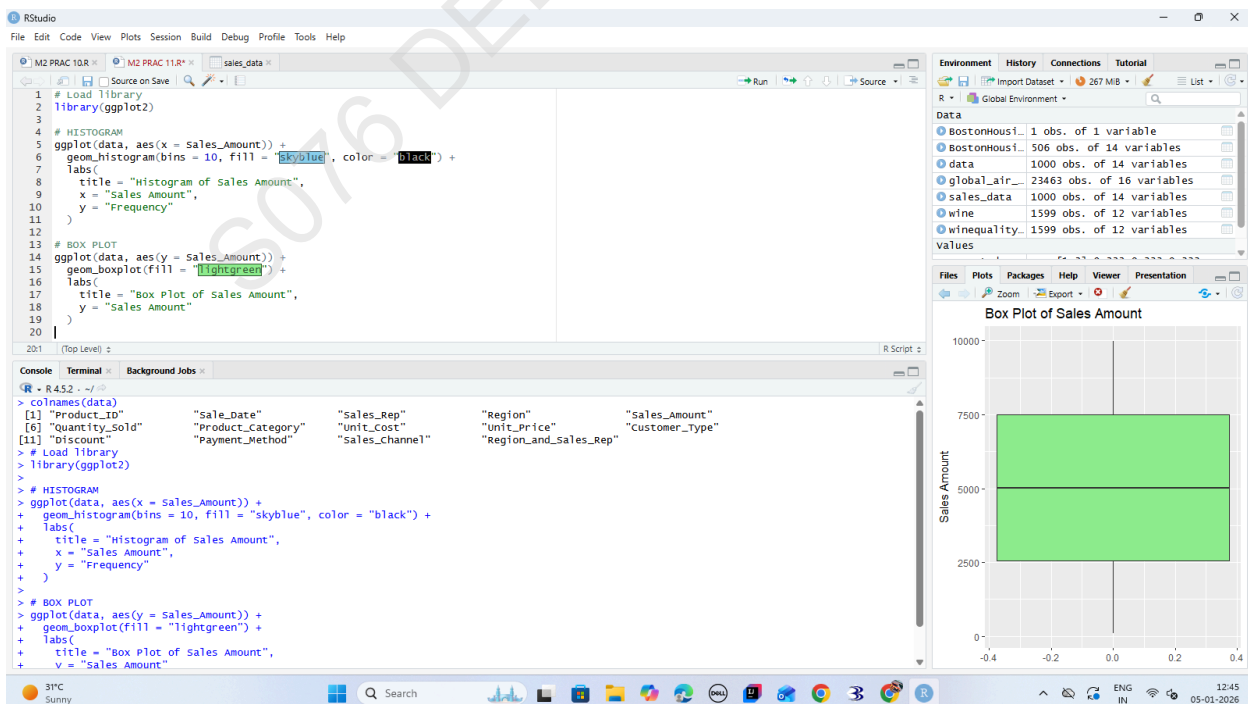
SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: DATA ANALYSIS WITH R

PRACTICAL NO. 10 - 12

AIM: 10 Creating graphical reports using ,ggplot2 (R).



11 Generating histograms and box plots using ggplot2 (R).



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SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: DATA ANALYSIS WITH R

12 Generating correlation matrices using cor() (R).

The screenshot displays the RStudio interface. The script editor on the left contains the following R code:

```
1 setwd("D:/S076")
2
3 sales <- read.csv("sales_data.csv")
4
5 print(sales)
6 str(sales)
7
8 sales_numeric <- sales[, apply(sales, is.numeric)]
9
10 print(sales_numeric)
11
12 cor_matrix <- cor(sales_numeric)
13 print(cor_matrix)
14
15 cor_complete <- cor(sales_numeric, use = "complete.obs")
16 print(cor_complete)
17
18 cor_spearman <- cor(sales_numeric, method = "spearman")
19 print(cor_spearman)
20
21 cor_sales_profit <- cor(sales[, c("Sales", "Profit")], use = "complete.obs")
22 print(cor_sales_profit)
23
```

The console on the right shows the output of the code, displaying the first 37 rows of the 'sales' data frame. The data includes columns for Product_ID, Sale_Date, Sales_Rep, Region, Sales_Amount, Quantity_Sold, Product_Category, Unit_Cost, Unit_Price, Customer_Type, and Discount.

Product_ID	Sale_Date	Sales_Rep	Region	Sales_Amount	Quantity_Sold	Product_Category	Unit_Cost	Unit_Price	Customer_Type	Discount
1	1052 03-02-2023	Bob	North	5053.97	18	Furniture	152.75	267.22	Returning	0.09
2	1093 21-04-2023	Bob	West	4384.02	17	Furniture	3816.39	4209.44	Returning	0.11
3	1015 21-09-2023	David	South	4631.23	30	Food	261.56	371.40	Returning	0.20
4	1072 24-08-2023	Bob	South	2167.94	39	Clothing	4330.03	4467.75	New	0.02
5	1061 24-03-2023	Charlie	East	3750.20	13	Electronics	637.37	692.71	New	0.08
6	1021 11-02-2023	Charlie	West	3761.15	32	Food	900.79	1106.51	New	0.21
7	1083 11-04-2023	Bob	West	618.31	29	Furniture	2408.81	2624.09	Returning	0.14
8	1087 06-01-2023	Eve	South	7698.92	46	Furniture	3702.51	3964.65	New	0.12
9	1075 29-06-2023	David	South	4223.39	30	Furniture	738.06	1095.45	New	0.05
10	1075 09-10-2023	Charlie	West	8239.58	18	Clothing	2228.35	2682.34	New	0.13
11	1088 16-11-2023	Eve	North	8518.45	13	Furniture	2440.11	2517.60	New	0.23
12	1100 14-08-2023	Bob	West	2198.74	43	Food	1100.81	1137.44	Returning	0.08
13	1024 11-11-2023	Eve	West	6607.80	21	Food	622.01	641.09	Returning	0.00
14	1003 31-12-2023	Alice	South	4775.59	30	Furniture	4190.28	4270.65	New	0.20
15	1022 17-08-2023	Charlie	South	8813.55	21	Food	2537.20	2869.60	New	0.29
16	1053 16-10-2023	Bob	North	2235.83	48	Furniture	121.19	487.65	New	0.18
17	1002 30-05-2023	David	North	6810.35	17	Furniture	4024.76	4420.15	Returning	0.04
18	1088 04-10-2023	Bob	East	6116.75	40	Electronics	4904.93	5034.35	New	0.10
19	1030 17-07-2023	David	West	3023.48	19	Clothing	3049.33	3209.22	Returning	0.26
20	1038 11-03-2023	Bob	South	1452.35	15	Clothing	2543.36	2790.10	Returning	0.07
21	1002 22-04-2023	Eve	North	6551.23	9	Electronics	4398.16	4439.12	New	0.18
22	1064 04-01-2023	Eve	East	7412.11	10	Electronics	4764.96	5074.42	New	0.12
23	1060 16-12-2023	Eve	East	3224.71	44	Clothing	3784.96	4276.99	New	0.06
24	1021 27-11-2023	Alice	South	6483.84	31	Furniture	2254.99	2441.79	Returning	0.24
25	1033 14-11-2023	David	South	4011.80	23	Furniture	2981.50	3360.40	Returning	0.22
26	1076 16-12-2023	Eve	East	7160.75	30	Clothing	3519.63	3774.65	New	0.20
27	1058 05-04-2023	Alice	North	2072.23	33	Furniture	1011.65	1084.28	Returning	0.07
28	1022 01-06-2023	David	East	8913.13	9	Electronics	2263.65	2558.95	New	0.03
29	1089 07-11-2023	Bob	West	2945.36	47	Furniture	4157.62	4314.56	New	0.17
30	1049 17-05-2023	Alice	West	3741.08	1	Furniture	3290.89	3317.75	Returning	0.17
31	1091 04-09-2023	Charlie	South	675.11	44	Food	2085.46	2406.58	Returning	0.06
32	1059 31-08-2023	Alice	North	1203.97	35	Food	3333.64	3764.52	Returning	0.02
33	1042 31-01-2023	Alice	North	5207.03	11	Food	635.20	814.14	Returning	0.02
34	1092 09-02-2023	David	West	2749.17	34	Furniture	2037.41	2238.65	New	0.30
35	1060 17-08-2023	Bob	East	8371.25	16	Clothing	3975.99	4422.59	New	0.24
36	1080 05-02-2023	Charlie	South	245.46	9	Electronics	1141.52	1550.19	New	0.28
37	1015 11-08-2023	Eve	East	3853.03	32	Furniture	970.51	1330.84	Returning	0.06
38	1062 06-01-2023	Eve	East	3439.72	15	Clothing	4756.55	4888.46	New	0.29

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[illegible]

The screenshot displays the RStudio interface with the following components:

- Source Panel:** Contains R code for data manipulation. The code defines a data frame with 14 variables (Product_ID, Sale_Date, Sales_Rep, Region, Sales_Amount, Quantity_Sold, Product_Category, Unit_Cost, Unit_Price, Customer_Type, Discount, Payment_Method, Sales_Channel, Region_and_Sales_Rep) and then creates a numeric version of the sales data.
- Console Panel:** Shows the output of the R code, including the structure of the data frame and the first few rows of the resulting data.
- Environment Panel:** Lists the objects in the R environment, including 'BostonHousing' and 'data', along with their dimensions and variable types.
- Plots Panel:** Currently empty.
- Files Panel:** Shows the file explorer with the project files.

The R code in the Source panel is as follows:

```
# R452 - D:\50761\
54 Bank Transfer Online West-Charlie
55 Bank Transfer Online West-David
56 Credit Card Online North-Alice
57 Cash Retail West-Alice
58 Credit Card Online East-David
59 Credit Card Retail East-David
60 Cash Online West-Alice
61 Bank Transfer Retail East-Charlie
62 Credit Card Online East-Eve
63 Bank Transfer Retail West-Eve
64 Bank Transfer Retail South-David
65 Cash Online North-Eve
66 Credit Card Online East-Alice
67 Credit Card Retail South-Alice
68 Bank Transfer Online East-David
69 Bank Transfer Online East-David
70 Cash Online North-David
71 Bank Transfer Retail East-Charlie
[ reached 'max' / getoption("max.print") -- omitted 929 rows ]

> str(sales)
'data.frame': 1000 obs. of 14 variables:
 $ Product_ID : int 1052 1093 1015 1072 1061 1021 1083 1087 1075 1075 ...
 $ Sale_Date : chr "03-02-2023" "21-04-2023" "21-09-2023" "24-08-2023" ...
 $ Sales_Rep : chr "Bob" "Bob" "David" "Bob" ...
 $ Region : chr "North" "West" "South" "South" ...
 $ Sales_Amount : num 5054 4384 4631 2168 3750 ...
 $ Quantity_Sold : int 18 17 30 39 13 32 29 46 30 18 ...
 $ Product_Category : chr "Furniture" "Furniture" "Food" "Clothing" ...
 $ Unit_Cost : num 153 3816 262 4330 637 ...
 $ Unit_Price : num 267 4209 371 4468 693 ...
 $ Customer_Type : chr "Returning" "Returning" "Returning" "New" ...
 $ Discount : num 0.09 0.11 0.2 0.02 0.08 0.21 0.14 0.12 0.05 0.13 ...
 $ Payment_Method : chr "Cash" "Cash" "Bank Transfer" "Credit Card" ...
 $ Sales_Channel : chr "Online" "Retail" "Retail" "Retail" ...
 $ Region_and_Sales_Rep : chr "North-Bob" "West-Bob" "South-David" "South-Bob" ...

> sales_numeric <- sales[, sapply(sales, is.numeric)]

> print(sales_numeric)
  Product_ID Sales_Amount Quantity_Sold Unit_Cost Unit_Price Discount
1 1052 5053.97 18 152.75 267.22 0.09
2 1093 4384.02 17 3816.39 4209.44 0.11
3 1015 4631.23 30 261.56 371.40 0.20
4 1072 2167.94 39 4330.03 4467.75 0.02
5 1061 2760.30 12 637.37 693.71 0.08
```

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Console Output (First Screenshot):

```
R - R 4.5.2 - D:\S076\
> $ Payment_Method : chr "Cash" "Cash" "Bank Transfer" "Credit Card" ...
> $ sales_channel : chr "online" "Retail" "Retail" "Retail" ...
> $ Region_and_Sales_Rep: chr "North-Bob" "West-Bob" "South-David" "South-Bob" ...
>
> sales_numeric <- sales[, apply(sales, is.numeric)]
> print(sales_numeric)
  Product_ID sales_Amount quantity_sold unit_cost unit_price discount
1      1052      5053.97          18      152.75      267.22      0.09
2      1093      4384.02          17      3816.39      4209.44      0.11
3      1015      4631.23          30       261.56       371.40      0.20
4      1072      2167.94          39      4330.03      4467.75      0.02
5      1061      3750.20          13       637.37       692.71      0.08
6      1021      3761.15          32       900.79      1106.51      0.21
7      1083       618.31          29      2408.81      2624.09      0.14
8      1087      7698.92          46      3702.51      3964.65      0.12
9      1075      4223.39          30       738.06      1095.45      0.05
10     1075      8239.58          18      2228.35      2682.34      0.13
11     1088      8518.45          13      2440.11      2517.60      0.23
12     1100      2198.74          43      1100.81      1137.44      0.08
13     1024      6607.80          21       622.01       641.09      0.00
14     1003      4775.59          30      4190.28      4270.65      0.20
15     1022      8813.55          21      2537.20      2869.60      0.29
16     1053      2235.83          48       121.19       487.65      0.18
17     1002      6810.35          17      4024.76      4420.15      0.04
18     1088      6116.75          40      4904.93      5034.35      0.10
19     1030      3023.48          19      3049.33      3209.22      0.26
20     1038      1452.35          15      2543.36      2790.10      0.07
21     1002      6551.23           9      4398.16      4439.12      0.18
22     1064      7412.11          10      4764.96      5074.42      0.12
23     1060      3224.71          44      3784.96      4276.99      0.06
24     1021      6483.84          31      2254.99      2441.79      0.24
25     1033      4011.80          23      2981.50      3360.40      0.22
26     1076      7160.75          30      3519.63      3774.65      0.20
27     1058      2072.23          33      1011.65      1084.28      0.07
28     1022      8913.13           9      2263.65      2558.95      0.03
29     1089      2945.36          47      4157.62      4314.56      0.17
30     1049      3741.08           1      3290.89      3317.75      0.17
31     1091      675.11          44      2085.46      2406.58      0.06
32     1059      1201.97          35      3333.64      3764.52      0.02
33     1042      5207.03          11       635.20       814.14      0.02
34     1092      2749.17          34      2037.41      2238.65      0.30
35     1060      8371.25          16      3975.99      4422.59      0.24
36     1080      245.46           9      1141.52      1550.19      0.28
```

Environment Pane (First Screenshot):

- BostonHousing: 506 obs. of 14 variables
- cor_complete: num [1:6, 1:6] 1 0.0102 0.09...
- cor_matrix: num [1:6, 1:6] 1 0.0102 0.09...
- cor_spearman: num [1:6, 1:6] 1 0.00923 0.0...
- data: 1000 obs. of 14 variables
- global_afr: 23463 obs. of 16 variables
- sales: 1000 obs. of 14 variables
- sales_data: 1000 obs. of 14 variables
- sales_numeric: 1000 obs. of 6 variables

Console Output (Second Screenshot):

```
R - R 4.5.2 - D:\S076\
156 1042 9442.65 20 3115.06 3565.58 0.30
157 1077 9605.34 12 2427.46 2743.09 0.29
158 1051 5262.45 23 3325.43 3401.10 0.10
159 1063 9775.35 4 1134.67 1201.39 0.30
160 1096 7597.37 23 608.57 1032.22 0.25
161 1052 1700.55 48 3002.35 3080.61 0.18
162 1096 4821.31 48 2449.21 2714.12 0.01
163 1004 7211.48 1 182.99 211.18 0.27
164 1094 2548.67 25 1933.39 1982.92 0.17
165 1023 6442.09 2 575.32 914.77 0.26
166 1015 6699.34 30 2680.22 3027.92 0.25
[ reached 'max' / getoption("max.print") -- omitted 834 rows ]
>
> cor_matrix <- cor(sales_numeric)
> print(cor_matrix)
  Product_ID sales_Amount quantity_sold unit_cost unit_price discount
Product_ID 1.00000000 0.010220945 0.09447851 0.013402371 0.01511639 -0.02209666
sales_Amount 0.01022095 1.000000000 -0.04159870 0.009769512 0.01181202 0.02315270
quantity_sold 0.09447851 -0.041598705 1.000000000 0.053887557 0.05729593 -0.00780626
unit_cost 0.01340237 0.009769512 0.05388756 1.000000000 0.99505556 -0.01774109
unit_price 0.01511639 0.011812024 0.05729593 0.995055560 1.000000000 -0.01702684
discount -0.02209666 0.023152703 -0.00780626 -0.017741095 -0.01702684 1.00000000
>
> cor_complete <- cor(sales_numeric, use = "complete.obs")
> print(cor_complete)
  Product_ID sales_Amount quantity_sold unit_cost unit_price discount
Product_ID 1.00000000 0.010220945 0.09447851 0.013402371 0.01511639 -0.02209666
sales_Amount 0.01022095 1.000000000 -0.04159870 0.009769512 0.01181202 0.02315270
quantity_sold 0.09447851 -0.041598705 1.000000000 0.053887557 0.05729593 -0.00780626
unit_cost 0.01340237 0.009769512 0.05388756 1.000000000 0.99505556 -0.01774109
unit_price 0.01511639 0.011812024 0.05729593 0.995055560 1.000000000 -0.01702684
discount -0.02209666 0.023152703 -0.00780626 -0.017741095 -0.01702684 1.00000000
>
> cor_spearman <- cor(sales_numeric, method = "spearman")
> print(cor_spearman)
  Product_ID sales_Amount quantity_sold unit_cost unit_price discount
Product_ID 1.000000000 0.009233241 0.093551269 0.01127903 0.01331226 -0.022929670
sales_Amount 0.009233241 1.000000000 -0.041112599 0.01023863 0.01313793 0.022751454
quantity_sold 0.093551269 -0.041112599 1.000000000 0.05280637 0.05600303 -0.006984497
unit_cost 0.011279030 0.010238626 0.052806369 1.000000000 0.99503751 -0.017926810
unit_price 0.013312264 0.013137931 0.056003035 0.99503751 1.000000000 -0.016805181
discount -0.022929670 0.022751454 -0.006984497 -0.01792681 -0.01680518 1.000000000
>
> sales_profit <- cor(sales$Sales, sales$Profit, use = "complete.obs")
```

Environment Pane (Second Screenshot):

- BostonHousing: 506 obs. of 14 variables
- cor_complete: num [1:6, 1:6] 1 0.0102 0.09...
- cor_matrix: num [1:6, 1:6] 1 0.0102 0.09...
- cor_spearman: num [1:6, 1:6] 1 0.00923 0.0...
- data: 1000 obs. of 14 variables
- global_afr: 23463 obs. of 16 variables
- sales: 1000 obs. of 14 variables
- sales_data: 1000 obs. of 14 variables
- sales_numeric: 1000 obs. of 6 variables

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